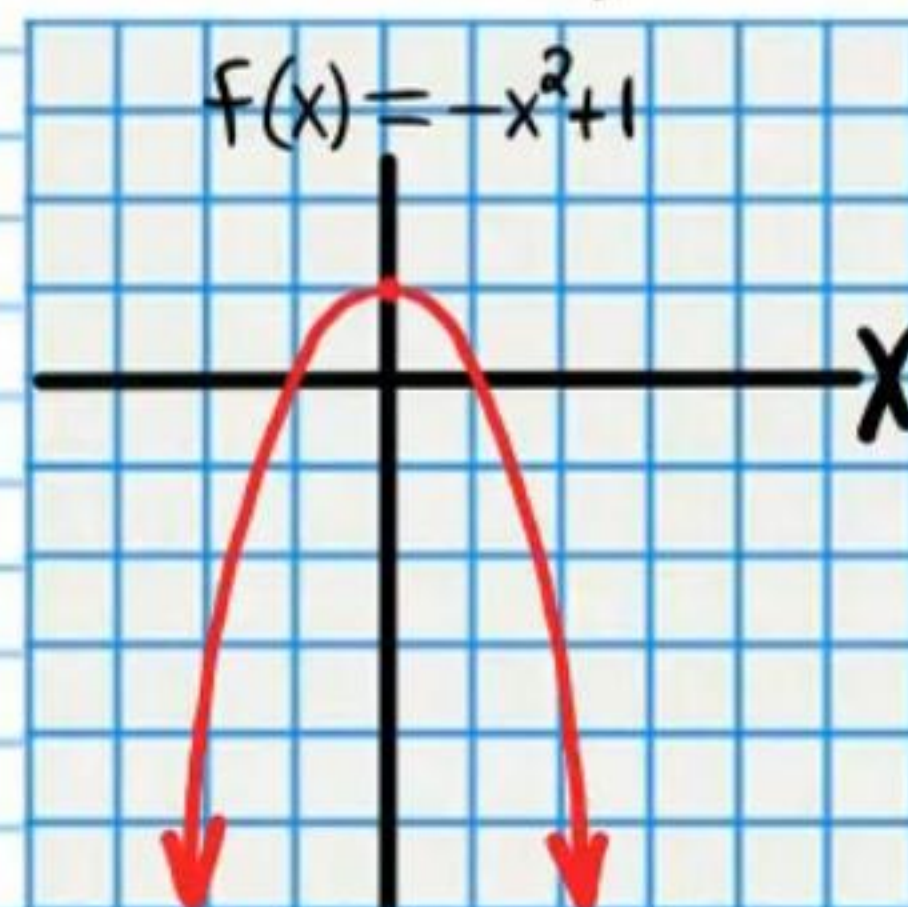


Name
Date
Course

The Derivative (An Introduction to Differential Calculus) Homework

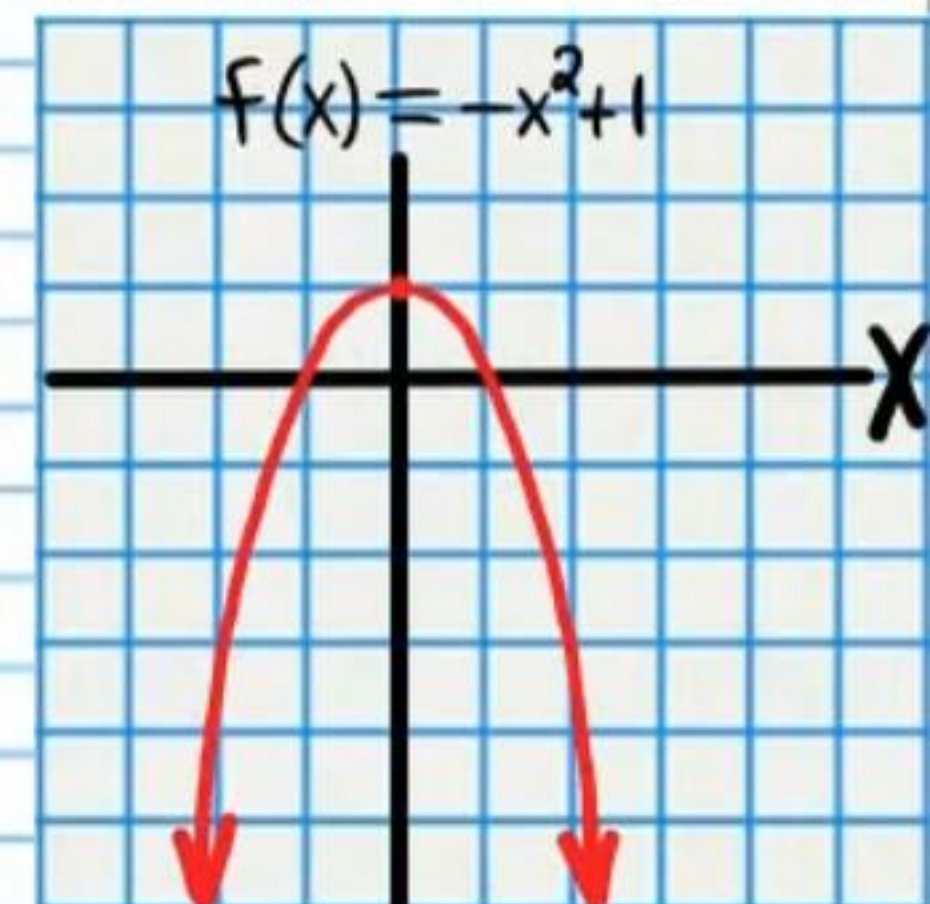
- ①^o) Given $f(x) = -x^2 + 1$,
Find $f'(x)$.

- ii) Draw the line tangent to the graph of $f(x)$ at $x=1$ on the image below.



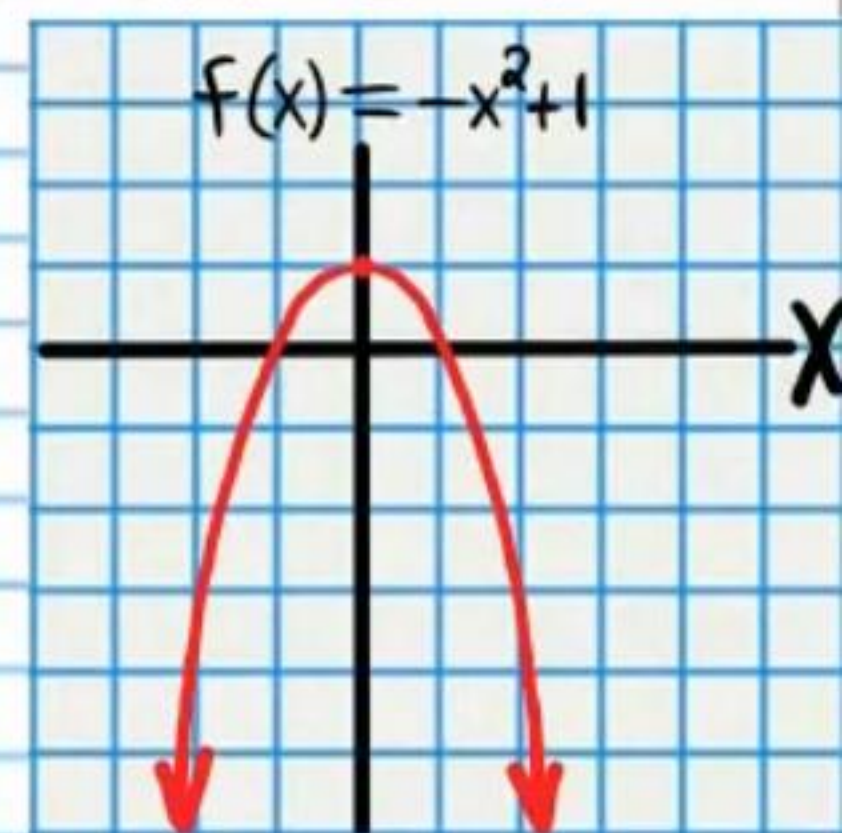
- iii) Find the slope of the line tangent to $f(x)$ at $x=1$.

- iv) Draw the line tangent to the graph of $f(x)$ at $x=-2$ on the image below.



- v) Find the slope of the line tangent to $f(x)$ at $x=-2$.

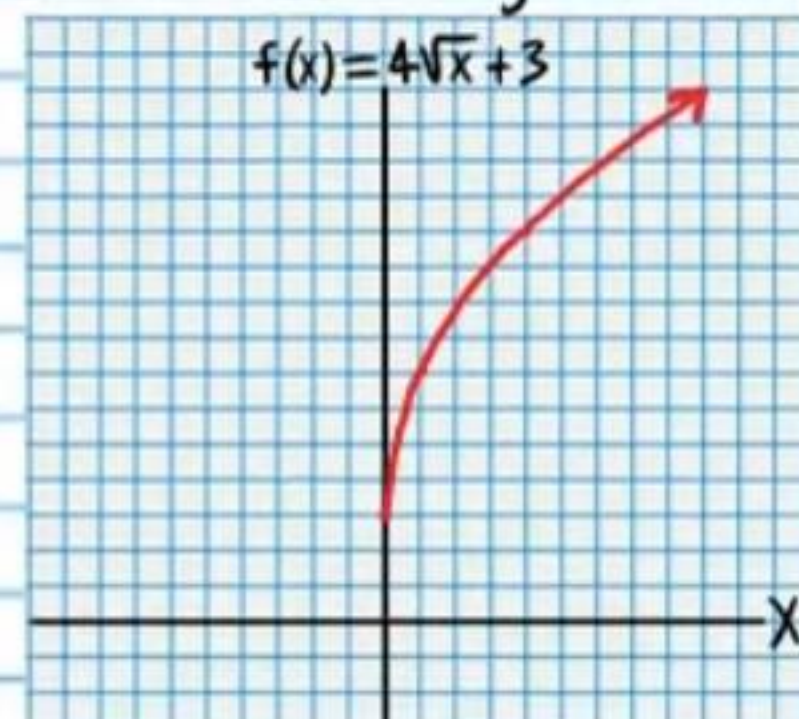
- vi) Draw the line tangent to the graph of $f(x)$ at $x=-1$ on the image below.



- vii) Find the slope of the line tangent to $f(x)$ at $x=-1$.

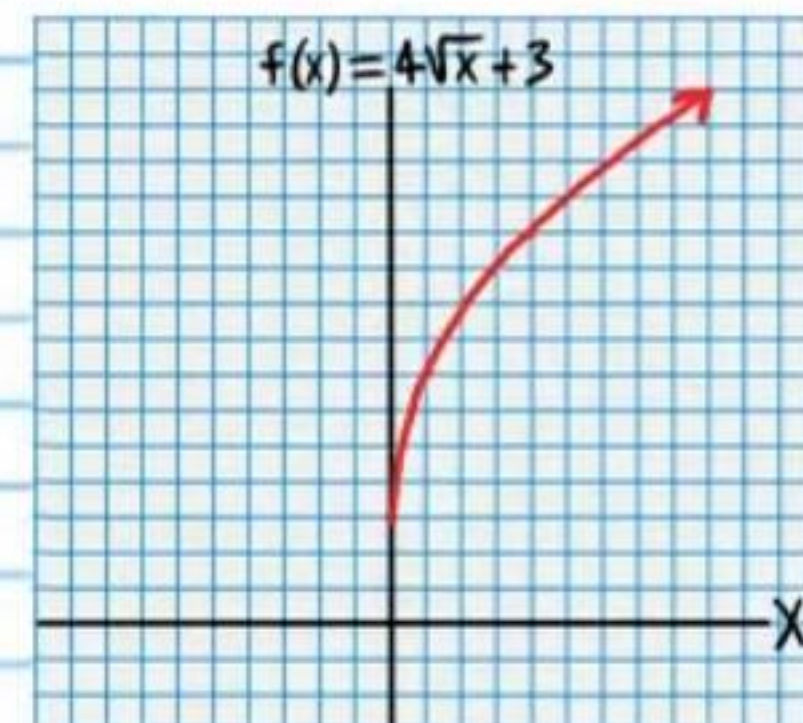
② i) Given $f(x) = 4\sqrt{x} + 3$,
Find $f'(x)$.

ii) Draw the line tangent
to the graph of $f(x)$ at
 $x=1$ on the image below.



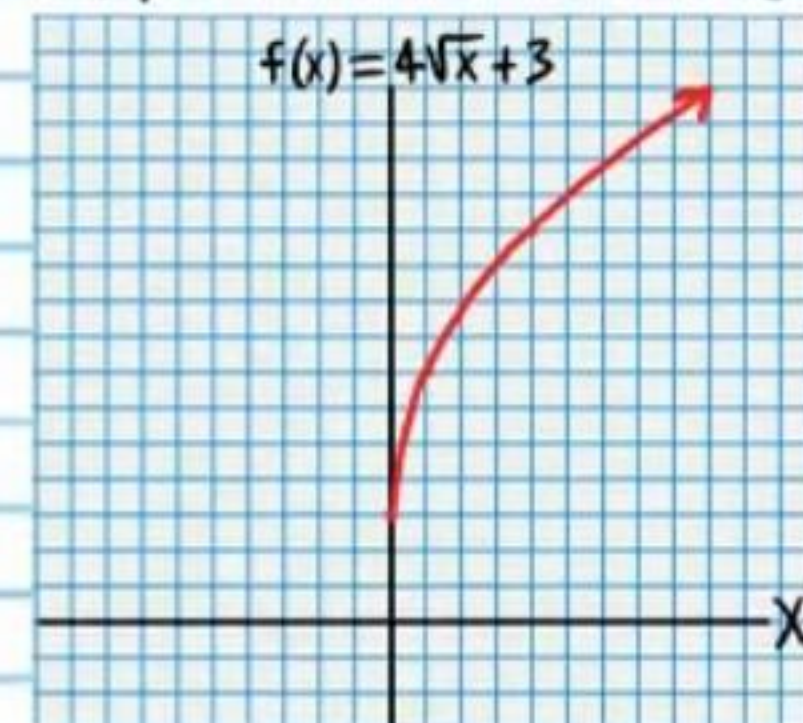
iii) Find the slope of the
line tangent to $f(x)$ at $x=1$.

iv) Draw the line tangent to the graph of $f(x)$ at $x=4$ on the
image below.



v) Find the slope of the line tangent to $f(x)$ at $x=4$.

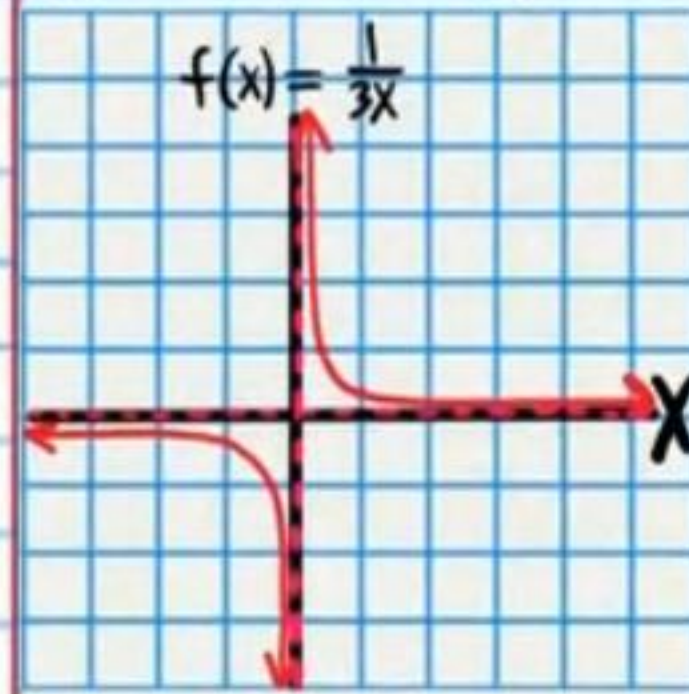
vi) Draw the line tangent to the graph of $f(x)$ at $x=5$ on the
image below.



vii) Find the slope of the line tangent to $f(x)$ at $x=5$.

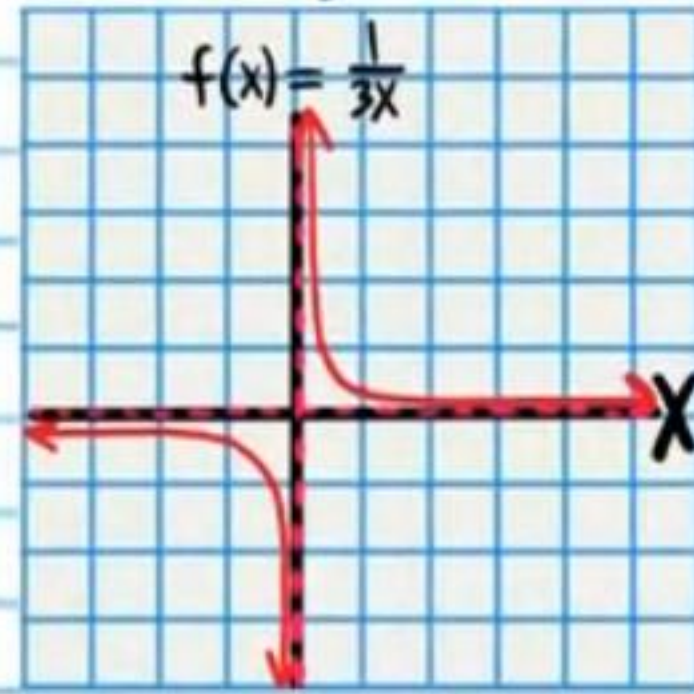
③ i) Given $f(x) = \frac{1}{3x}$, find $f'(x)$.

ii) Draw the line tangent to the graph of $f(x)$ at $x = -1$ on the image below.



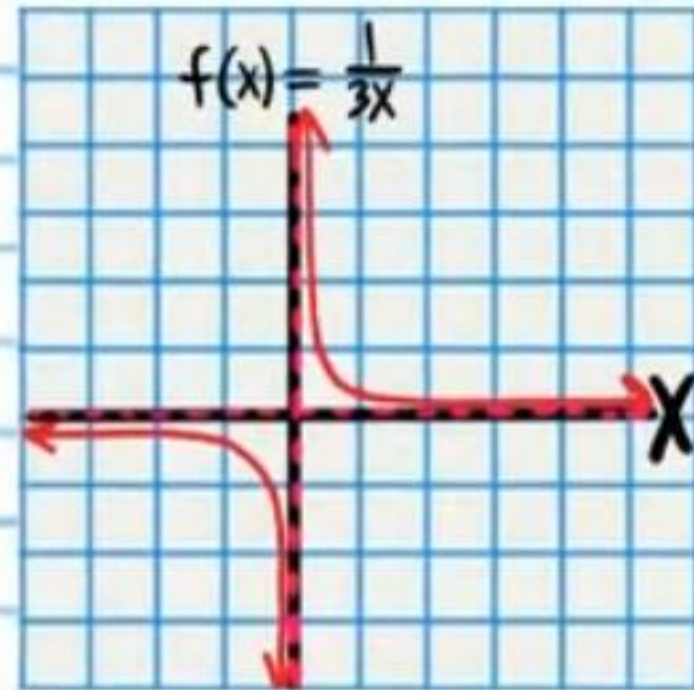
iii) Find the slope of the line tangent to $f(x)$ at $x = -1$.

iv) Draw the line tangent to the graph of $f(x)$ at $x = 2$ on the image below.



v) Find the slope of the line tangent to $f(x)$ at $x = 2$.

vi) Draw the line tangent to the graph of $f(x)$ at $x = 1$ on the image below.



vii) Find the slope of the line tangent to $f(x)$ at $x = 1$.

④ i) Given $f(x) = x^2 + x - 10$
Find $f'(x)$.

ii) Find the instantaneous
rate of change of $f(x)$
at $x = -3$.

iii) Find the instantaneous
rate of change of $f(x)$
at $x = 1\frac{2}{3}$.

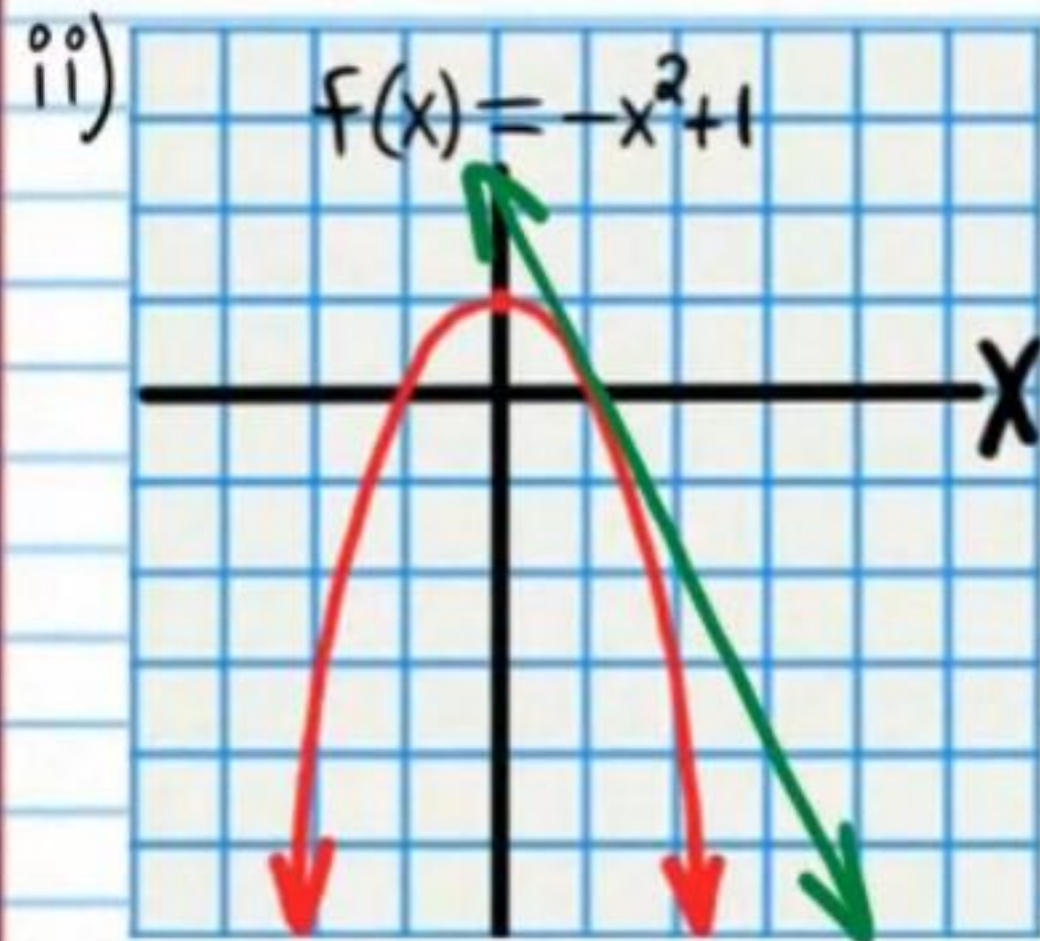
⑤ i) Given $f(x) = -\frac{1}{4}x + 6$
Find $f'(x)$.

ii) Find the instantaneous
rate of change of $f(x)$
at $x = \frac{4}{5}$.

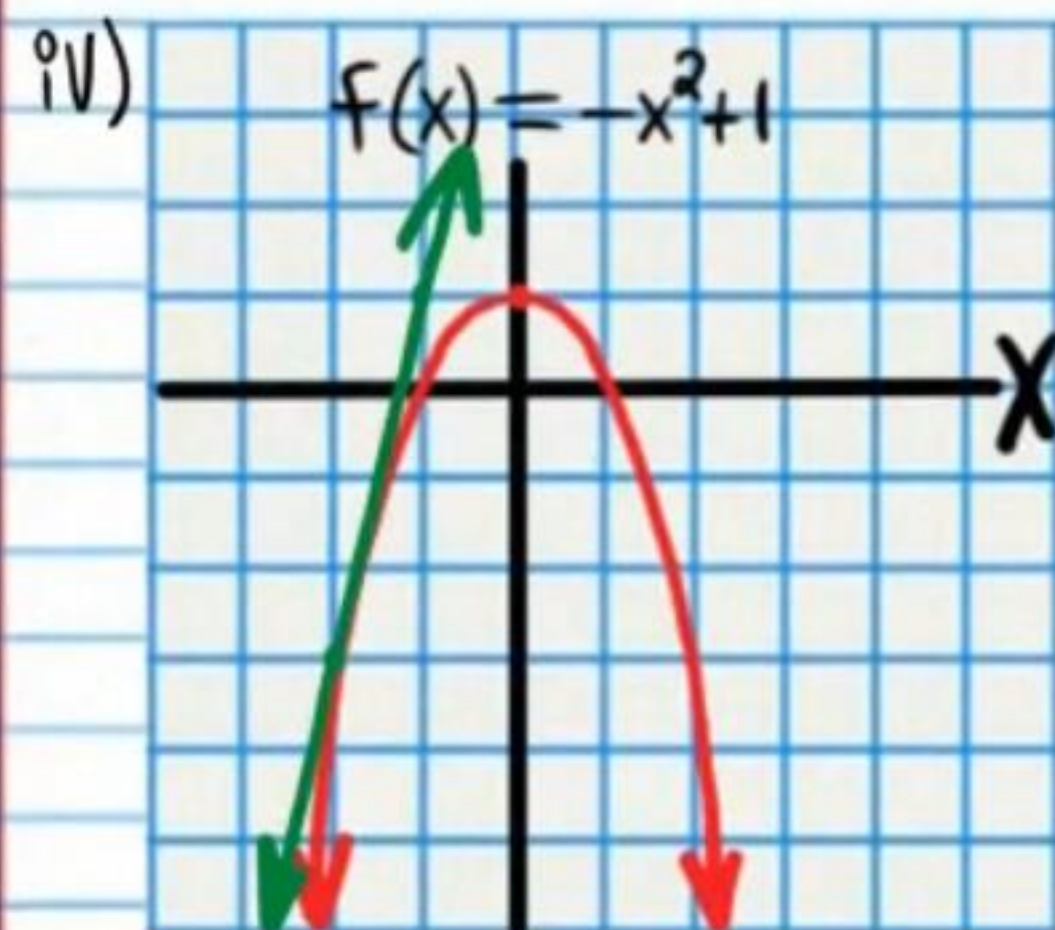
iii) Find the instantaneous
rate of change of $f(x)$
at $x = 6.5$.

Homework Answers

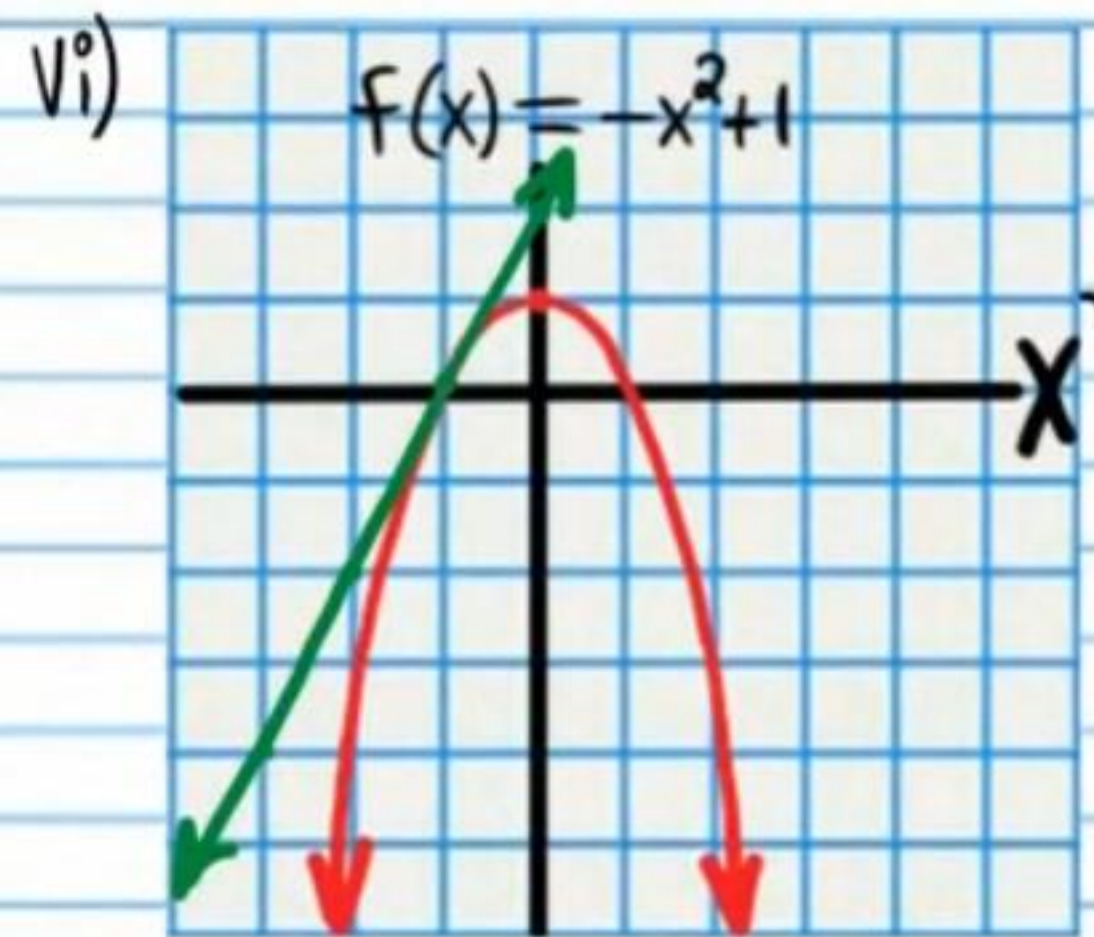
① i) $f'(x) = -2x$



iii) -2

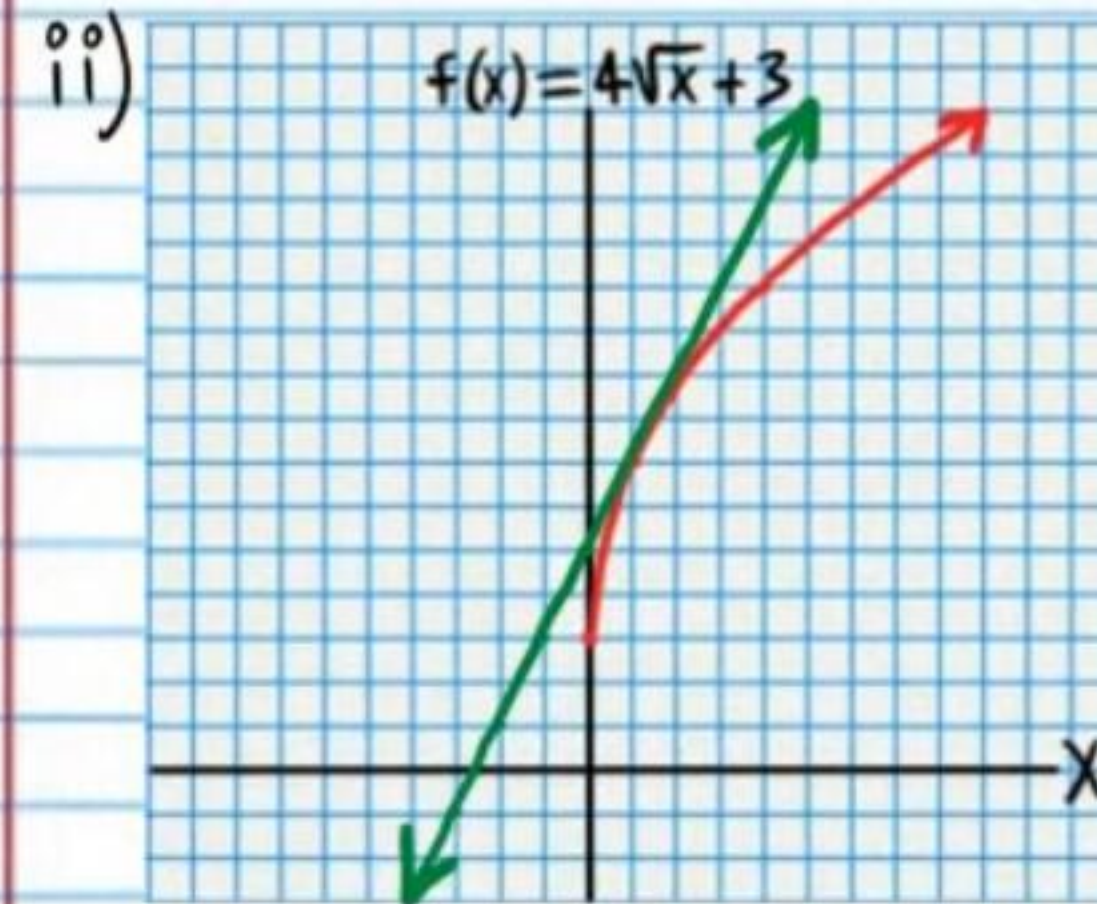


v) 4

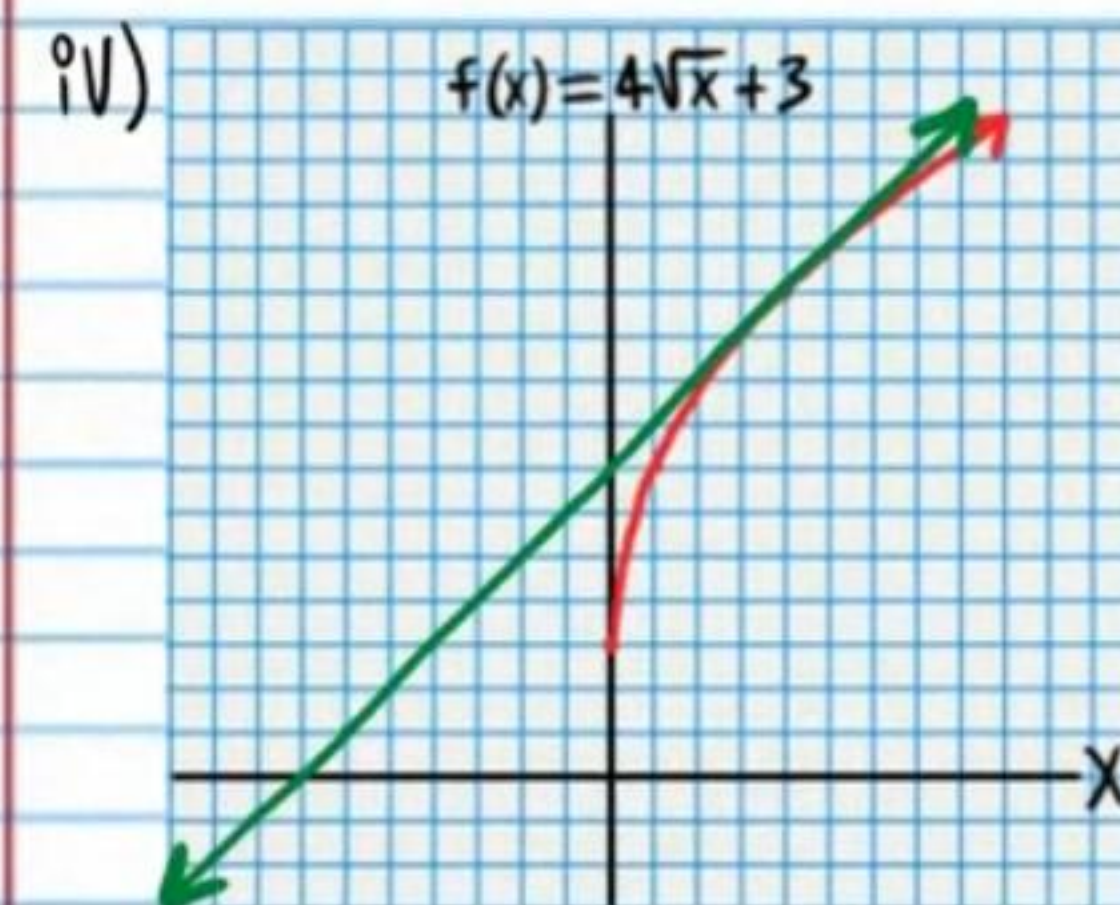


vii) 2

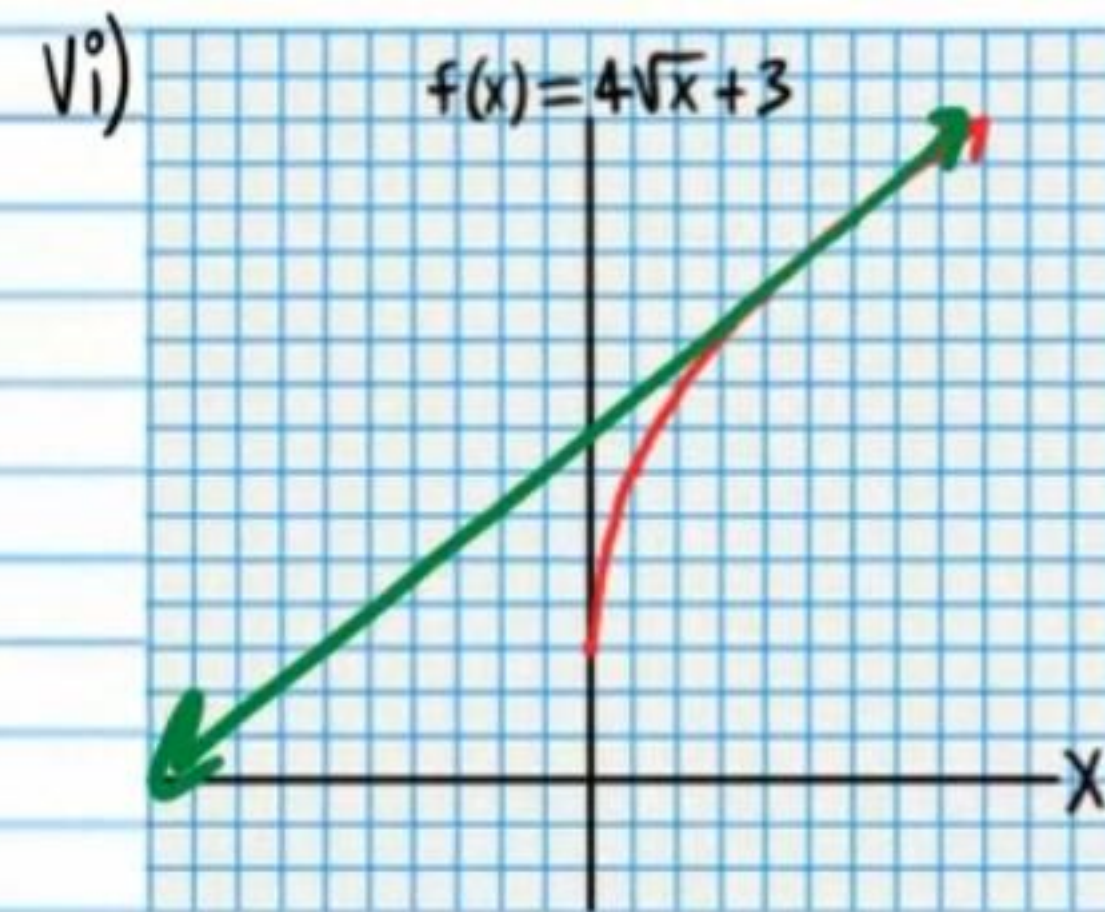
② i) $f'(x) = \frac{2}{\sqrt{x}}$



iii) 2

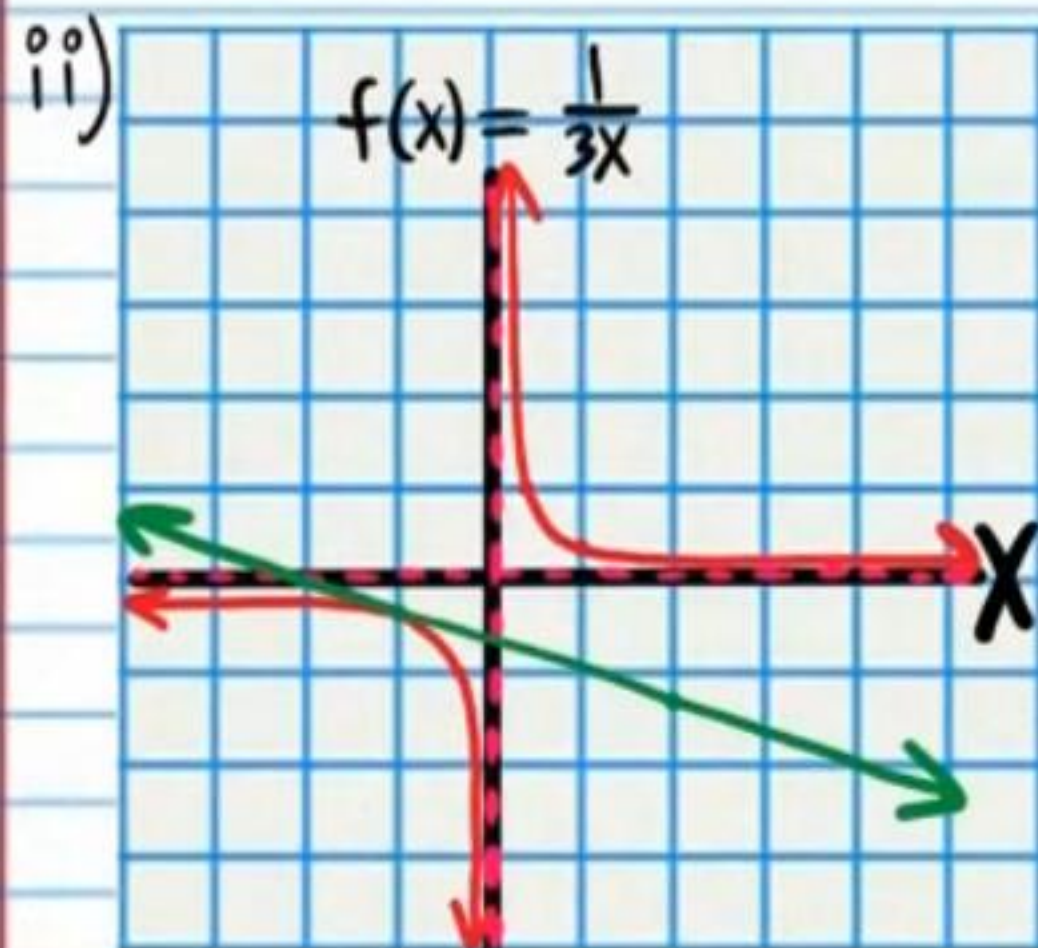


v) 1

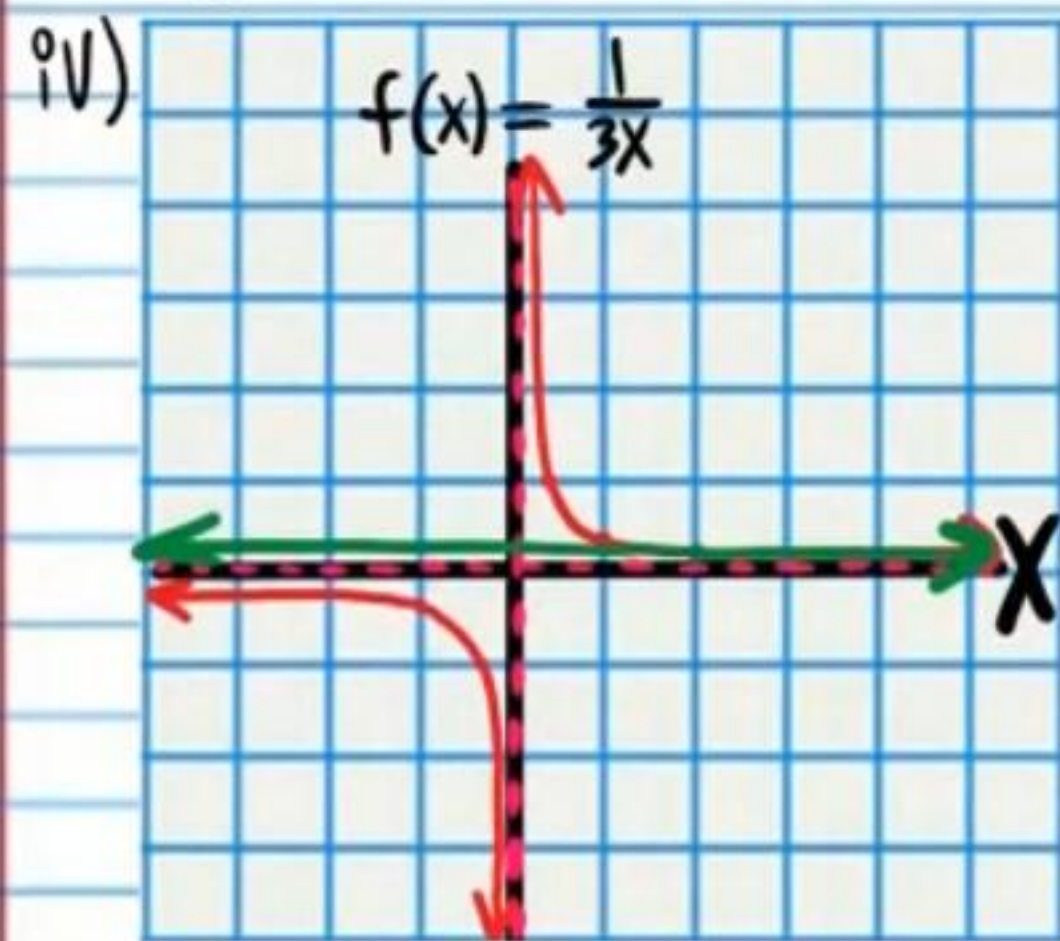


vii) $\frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$

③ i) $f'(x) = -\frac{1}{3x^2}$

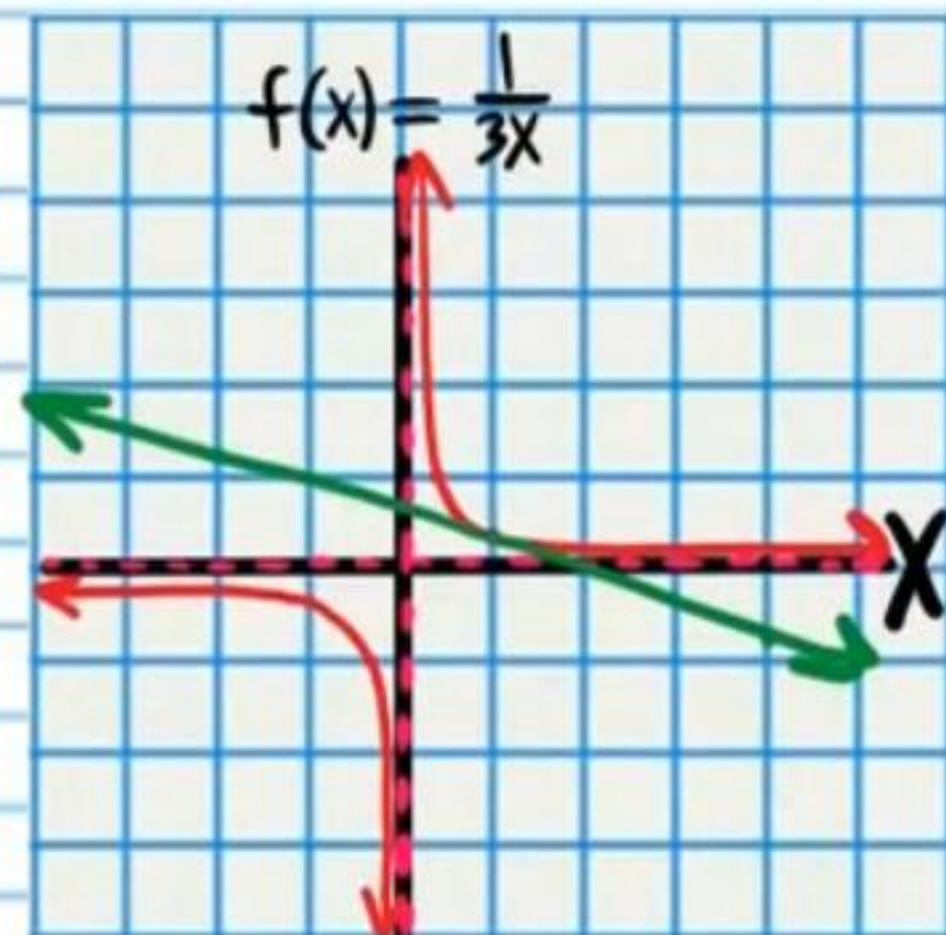


iii) $-\frac{1}{3}$



v) $-\frac{1}{12}$

vi)



vii) $-\frac{1}{3}$

④ i) $f'(x) = 2x + 1$

ii) -5

iii) $\frac{13}{3} = 4\frac{1}{3}$

⑤ i) $f'(x) = -\frac{1}{4}$

ii) $-\frac{1}{4}$

iii) $-\frac{1}{4}$